

Exercise 1: Linear Regression I

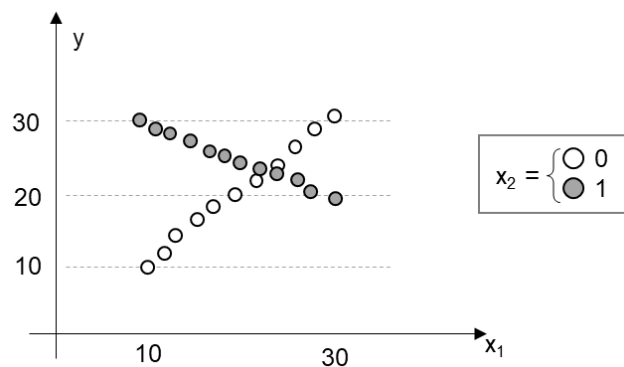
Consider the following three points in an (x, y) plane: $(2, 2)$, $(3, 4)$, and $(8, 5)$.

- a) Using a solver (e.g. in MS Excel), compute the best regression line $y = a + bx$. Based on the actual and predicted values of this regression, what is the TSS, the ESS, and – based on these – the R^2 value?

See Excel

Exercise 2: Linear Regression II

The following figure shows a schematic scatter plot of data points, the y -values of which are to be explained by different linear regression models based on x_1 and x_2 .



- a) In the following table, please indicate the coefficients for the respective regression model estimates (grey fields do not exist in the respective model).

	Model I	Model II	Model III	Model IV	Model V
Intercept	22.5	17.5	20	15	0
x_1		0.25		0.25	1
x_2			5	5	35
$x_1 * x_2$					-1.5

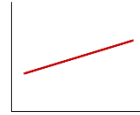
Model I

- Model can only be a straight line with now slope -> average value of points
- Half of the dots are between 10 and 30 (20 on avg)
- The other half is between 30 and 20 (25 on avg)
- 22.5 is average across both lines



Model II

- Model has an intercept and a slope dependent on x_1
- Starting at 20 (the midpoint on the left), ending at 25 (the midpoint on the right)
-> slope = $5/20 = 0.25$ (or average of the white points slope of 1 and black points slope of -0.5)
- Y-offset = $20 - 10 \cdot 0.25$, or from central point $(20, 22.5) - 22.5 - 20 \cdot 0.25$



Model III

- Model has an intercept and can shift in the intercept dependent on x_2
- Intercept = Average of white points (20)
- X_2 = Distance between average of all grey points (25) and white points (20) = 5



Model IV

- Average slope is $(-0.5 + 1)/2 = 0.25$
- Anchor two parallel lines with slope 0.25 to each set of dots in the middle
- White: (20,20) → y-offset is $20 - 20 \cdot 0.25 = 15$
- Grey: (20,25) → y-offset is $25 - 20 \cdot 0.25 = 20$ → delta is $20 - 15 = 5$



Model V

- Fit the two lines perfectly
- X_1 : slope=1, offset=0
- X_2 : difference at $x_1=0$: which is 35
- $X_1 * X_2$: how do you have to change slopeWhite (x_1) to get to slopeGrey (x_2): -1.5



Exercise 3: Spline Regression

Consider the dataset *splines.csv*, which shows annual income as well as the age of participants measured during a study.

- a) Analyze the relationship between income and age using linear, quadratic, as well as spline regression models.

See R-Script

- b) Assuming retirement affects the annual income of participants, estimate the retirement age based on the dataset.

See R-Script

Exercise 4: Logarithmic transformation

Consider the following small dataset of apartment prices in Berlin (prices in thousands of EUR, sizes in square meters). We will explore how a log transformation affects the shape

and skewness of data distribution, outliers, variance and the interpretation of linear regression.

Apartment #	1	2	3	4	5	6	7	8	9	10	11	12
size (sqm)	40	55	60	70	80	90	100	120	140	160	180	220
price (kEUR)	276.7	249.8	309.1	479.8	488.1	264.7	469.2	417.9	485.3	622.8	649	1659

- a) Compute the **mean**, **median** and **skewness** for price and the natural logarithmic transformation of price. Visualize the values and interpret the results: How does the transformation affect the shape and spread of the distribution?

➔ See Excel. LN compresses large values, reduces right-skew/outlier impact, and often stabilizes variance.

- b) Fit two simple linear regression models and calculate the estimated coefficients and R^2 for both:

Model A: price = $\alpha_A + \beta_A \cdot \text{size}$

Model B: $\ln(\text{price}) = \alpha_B + \beta_B \cdot \text{size}$

Briefly compare: Which model explains more variance? Which seems less influenced by the extreme price?

➔ See Excel. Model B explains more variance and is less dominated by the extreme 220 sqm / 1658.8 kEUR point. Also, for apartments smaller than approx. 14 sqm Model A estimates a negative Price (kEUR), whereas Model B only estimates positive values for $\ln(\text{Price (kEUR)})$.

- c) Using **Model B**, interpret β_B in % terms: "Holding everything else constant, each additional 1 sqm increases price by about ____%". What are the approximate % changes in price for an additional 10 sqm and 20 sqm?

➔ See Excel. With $\log(\text{price}) = \gamma + \beta_B \cdot \text{size}$, a 1-sqm increase multiplies price by e^{β_B} . Here, $e^{0.008177} - 1 \approx 0.821\%$ per **additional 1 sqm**.

- d) When would you use a logarithmic transformation of price instead of price directly? When is that not appropriate?

➔ **Use $\log(y)$ when:** y is strictly positive; distribution is right-skewed; variance grows with the mean; effects are multiplicative/percentage-like.

➔ **Avoid/adjust when:** values can be 0 or negative (you'd need a shift like $\log(y + c)$, chosen and justified carefully); interpretability on the original scale is crucial and stakeholders expect additive changes.